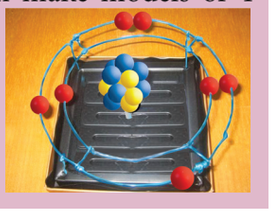
Activity 3

Look at the model given below. Make groups of five. Each group can make models of 4 elements by using available materials like balls, beads, string etc.



11.6 Isotopes, Isobars and Isotones

11.6.1 Isotopes

In nature, a number of atoms of elements have been identified, which have the same atomic number but different mass numbers. For example, take the case of hydrogen atom, it has three atomic species as shown below:

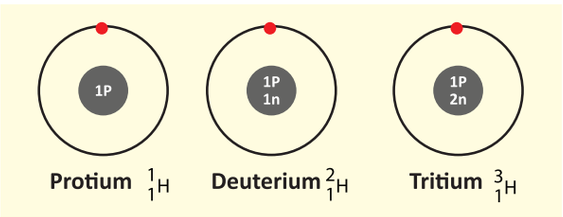


Figure 11.9 Isotopes

The atomic number of all the three isotopes is 1, but the mass number is 1, 2 and 3, respectively. Other such examples are: i) carbon, $^{12}_{6}C$, $^{13}_{6}C$ ii) Chlorine $^{35}_{17}Cl$ $^{37}_{17}Cl$

On the basis of these examples, isotopes are defined as the different atoms of the same element, having same atomic number but different mass numbers. There are two types of isotopes: **stable** and **unstable**. The isotopes which are unstable, as a result of the extra neutrons in their nuclei are radioactive and are called **radioisotopes**. For example, uranium-235, which is a source of nuclear reactors, and cobalt-60, which is used in radiotherapy treatment are both radioisotopes.

Activity 4

Draw the structures of the isotopes of oxygen O^{16} and O^{18}

Atomic number of oxygen = 8

11.6.2 Isobars

Let us consider two elements – calcium (atomic number 20), and argon (atomic number 18).

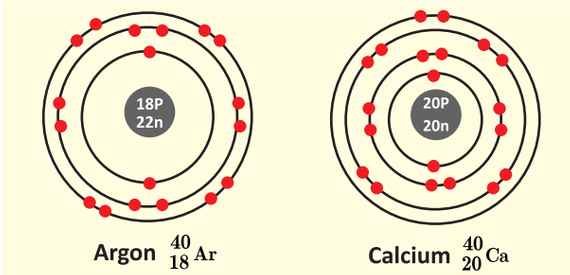


Figure 11.10 Isobars

They have (Fig. 11.10) different number of protons and electrons. But, the mass number of both these elements is 40. It follows that the total number of nucleons in both the atoms are the same. They are called isobars. Atoms of different elements with different atomic numbers, and same mass number are known as isobars.

More to Know

Thumb rule for isotopes and isobars. Remember **t** for top and **b** for bottom.
Isotope: Top value changes – Atomic mass
Isobars: Bottom value changes – Atomic number

11.6.3 Isotones

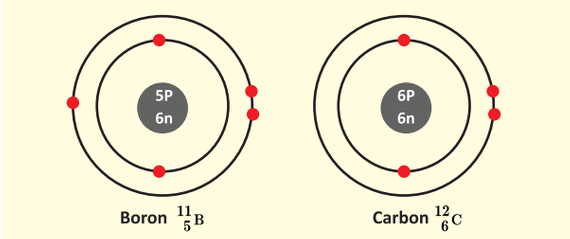


Figure 11.11 Isotones

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